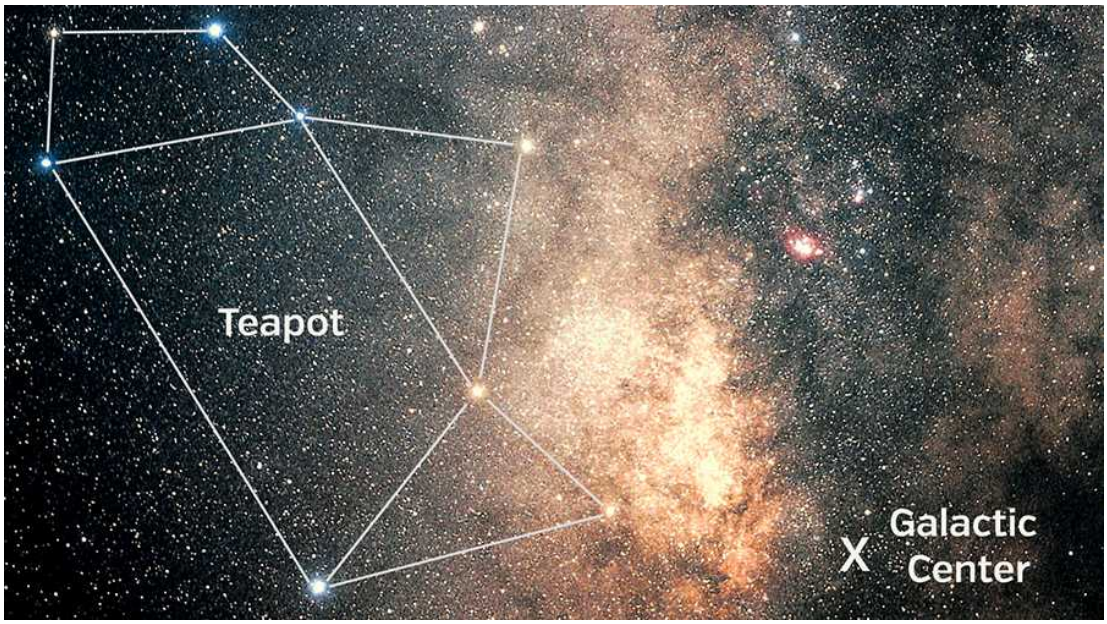


Fig. 5.9. The location of our galaxy's center on the sky. A giant black hole resides there.

Inside our own galaxy, there are roughly 100 million smaller black holes: holes that typically are between about three and thirty times as heavy as the Sun. We know this not because we've seen evidence for all these, but because astronomers have made a census of heavy stars that will become black holes when they exhaust their nuclear fuel. From that census, astronomers have inferred how many such stars have already exhausted their fuel and become black holes.

So black holes are ubiquitous in our universe. Fortunately, there are none in our solar system. If there were, the hole's gravity would wreak havoc with the Earth's orbit. The Earth would be thrown close to the Sun where it boils, or far from the Sun where it freezes, or even out of the solar system or into the black hole. We humans would survive for no more than a year or so!

Astronomers estimate that the nearest black hole to Earth is roughly 300 light-years away: a hundred times farther than the nearest star (other than the Sun), Proxima Centauri.